



CUBAN NATIONAL CHEMISTRY CONTEST 2022

"You should never be afraid of what you're doing, when it's right."

Marie Curie

Name(s): _____ Last name(s): _____

School: _____

Municipality: _____ Province: _____

CODE: _____ Test: **DAY 2. TWELFTH GRADE EXAM**

Instructions

- Write clearly all your data in the appropriate place (names, last names, school, municipality, province). Do not write anything in the field CODE.
- Verify that your questions booklet is complete. This is the Day 2 problem set, with an individual exam for 12th grade. The exam has 3 problems (problem 8, 4 items; problem 9, 7 items; problem 10, 2 items) and a total of 5 pages.
- All data, constants, conversions and basic equations needed are included in the exam. You are allowed to use a scientific calculator.
- At the beginning of each problem its scoring (from a total of 100 points), each item scoring (in marks) is indicated.

Success!



Ministerio de Educación
República de Cuba



CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Equation of ideal gas	$p \cdot V = n \cdot R \cdot T$
Gases constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	First Principle of Thermodynamics	$Q = \Delta E + W$
Zero on Celsius scale	273,15 K	Gibbs' free energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Ionic product of water at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp\left(-\frac{E_A}{R \cdot T}\right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$	$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF ELEMENTS

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1																	2
1 H 1.008	2											13	14	15	16	17	2 He 4.003
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3	4	5	6	7	8	9	10	11	12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.63	33 As 74.92	34 Se 78.97	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.95	43 Tc -	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57-71	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po -	85 At -	86 Rn -
87 Fr -	88 Ra -	89-103	104 Rf -	105 Db -	106 Sg -	107 Bh -	108 Hs -	109 Mt -	110 Ds -	111 Rg -	112 Cn -	113 Nh -	114 Fl -	115 Mc -	116 Lv -	117 Ts -	118 Og -

Problem 8. Electrolytic solutions (19 points)

8.1	8.2	8.3	8.4	Total	Total
4	3	7	5	19	19

Sodium hypochlorite is one of the most widely used substances for several centuries as a water disinfectant and bleach. In the process of obtaining hypochlorite, first, by means of electrolysis of a brine solution (NaCl), gaseous chlorine was produced. For this, a current of 11.5 A was passed for 2 h and 50 min to a volume of 1L thereof.

8.1 State the possible cathodic and anodic half-reactions, as well as the global equation of the process that occurs.

8.2 How much amount of substance of the chlorine substance is produced?

Chlorine that was obtained after the electrolysis was thoroughly bubbled into a 1 mol/L sodium hydroxide solution. The container where said solution was found, after adding all the chlorine, was hermetically closed.

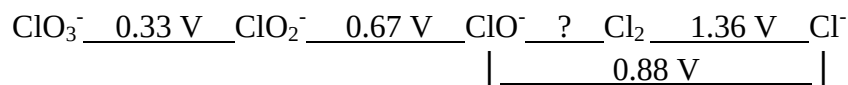
8.3 Show by calculations that the disproportionation of chlorine in a basic medium is a spontaneous process and state the global equation of this reaction.

8.4 Calculate the pH of the final solution. Note that the limiting substance is NaOH.

Data:

$$K_c(\text{H}_2\text{O}) = 1.86 \text{ K} \cdot \text{kg} \cdot \text{mol}^{-1}$$

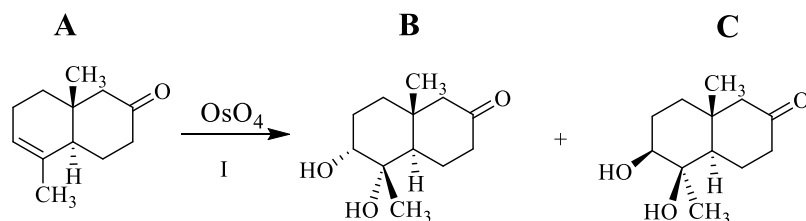
$$K_a(\text{HClO}) = 4.0 \cdot 10^{-8}$$



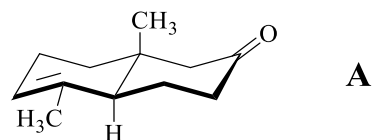
Problem 9. Cauhtemone (19 points)

9.1	9.2	9.3	9.4	9.5	9.6	9.7	Total	Total
2	5	16	9	2	2	2	38	19

Cauhtemone is a plant growth inhibitor isolated from the Mexican medicinal plant Cauhtematl. It can be obtained through a sequence of non-enantioselective reactions, whereby in each synthesis step the optically active compounds are obtained as mixtures of enantiomers. To simplify the scheme only one of the enantiomers per compound is shown. Starting from compound **A**, the first step of the synthesis is shown below:

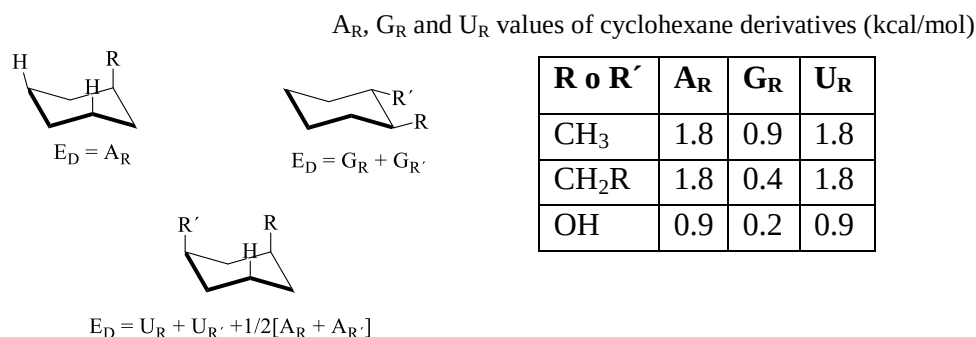


The three-dimensional structure of shown **A** stereoisomer can be represented in the chair form as follows:

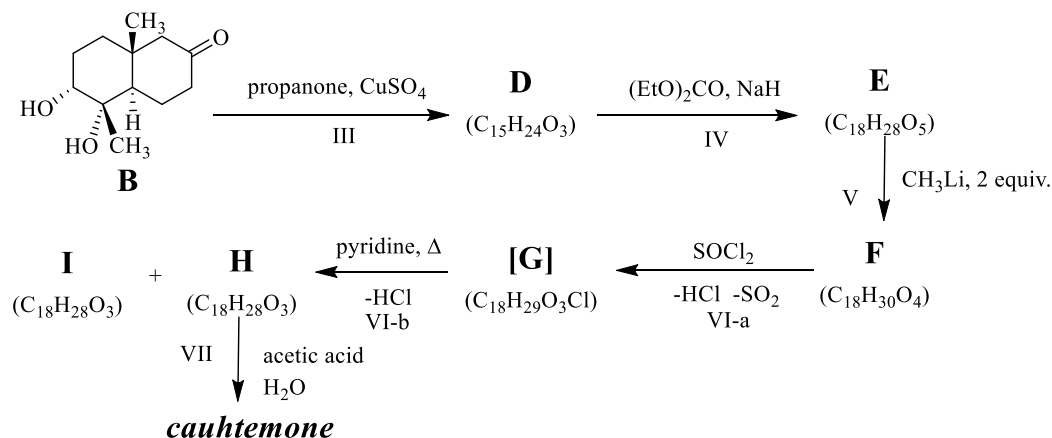


9.1 Illustrate the chair-shaped structure of **B** and **C**.

9.2 Justify by calculations which of the two diastereomers (**B** or **C**) is the most stable. Calculate the difference between the destabilization energies of diastereomers **B** and **C** using the data given in the following table. The destabilization energy (E_D) is the sum of the 1,3-diaxial interactions between a hydrogen atom and the R substituent (A_R), gauche interactions between the R and R' substituents (G_R), and 1,3- diaxial between the R and R' substituents (U_R).



The second stage of the synthesis of *cauhtemone*, starting from isomer **B**, proceeds as follows:



Steric effects determine that during the condensation of compound **D** with ethyl carbonate in the presence of sodium hydride, only the least hindered isomer is formed. The compound formed in the above reaction exists in the form of two tautomers, **T** and **E**, which are in equilibrium. Tautomer **E**

possesses intramolecular hydrogen bonding and is more stable than its tautomer **T**. In fact, the reaction between **D** and ethyl carbonate first forms the **T** tautomer (not shown in the diagram) which rapidly transforms into **E**. Treatment of compound **F** with thionyl chloride produces an intermediate compound **G** which, in the presence of base, rapidly transforms into compounds **H** and **I**, where **H** is the major product.

Catalytic hydrogenation of **cauhtemone** produces a dihydroderivative, whereas only a compound **Y** ($C_{12}H_{18}O_4$) and propanone are obtained by reductive ozonolysis of this natural product. Treating **cauhtemone** with 2,4-dinitrophenylhydrazine in acidic solution yields a product with a single hydrazone unit.

9.3 Write the three-dimensional structure of compounds **E** through **I**, **Y**, and **cauhtemone**. Mark the stereogenic centers in the **cauhtemone** structure and classify them according to the Cahn-Ingold-Prelog rules.

9.4 Write the mechanism of reactions IV, VI-a and VI-b.

9.5 Represent the tautomeric equilibrium between **E** and **T**.

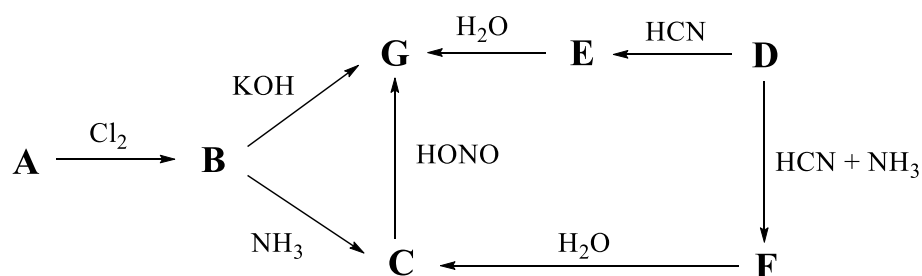
9.6 Why is **H** the major product in reaction VI-b?

9.7 State a reasonable sequence of steps for the synthesis of the reagent 2,4-dinitrophenylhydrazine from benzene.

Problem 10. Organic compounds (12 points)

10.1	10.2	Total	Total
8	16	24	12

Substance **G** can be prepared by various methods, according to the following scheme:



Compound **A** is 48.60% by mass carbon, 8.10% hydrogen, and 43.30% oxygen. When doing the volumetric titration of 0.222 g of **A** dissolved in water, 15.0 mL of 0.2 mol/L NaOH were needed for its total neutralization.

Compound **D** contains 54.54% carbon, 9.09% hydrogen, and 36.37% oxygen. It combines equimolarly with $NaHSO_3$ to give a compound with 21.6% sulfur by mass.

10.1 Write the empirical formula of **A** and **D**.

10.2 Deduce the structures of the compounds from **A** to **G**.